

$$\Rightarrow k=4$$

$$2x + 3y = 6$$

$$4x + 6y = 3k$$

$$\begin{bmatrix} 2 & 3 & 6 \\ 4 & 6 & 3k \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 6 \\ 0 & 0 & 3k-12 \end{bmatrix} \left. \vphantom{\begin{bmatrix} 2 & 3 & 6 \\ 0 & 0 & 3k-12 \end{bmatrix}} \right\} \text{Rank} = 2$$

$$\Rightarrow 3k - 12 = 0$$

$$\Rightarrow k=4$$